YOONG Khai Jie, Nathaniel

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PROFESSIONAL SUMMARY

FPGA and Firmware Engineer with 2+ years full-time at DSO National Laboratories, building on FPGA and hardware internships there since 2018. Designs and implements RTL (VHDL) for continuous, high-throughput data acquisition and DSP streaming pipelines on Xilinx FPGAs, using AXI4-Stream with buffering and backpressure, and AXI DMA into DDR memory. Closes timing on own designs, brings up and debugs new hardware in the lab, and writes embedded C for the processing side. Keen to grow FPGA design skills in a low-latency, high-throughput environment.

RELEVANT SKILLS

- **FPGA Design & Architecture:** RTL design and architecture (VHDL), AXI4-Stream, DSP streaming pipelines, buffering and backpressure, continuous high-throughput acquisition, AXI DMA + DDR
- **Implementation & Timing:** Xilinx Vivado, Vitis, Vitis HLS; synthesis, place & route, static timing analysis, timing closure (analysing timing reports, fixing failing paths in RTL)
- **Bring-up, Debug & Verification:** Lab bring-up of new hardware, oscilloscope, JTAG debugger, Vivado Simulator (XSim)
- **Memory & Interfaces:** DDR, QSPI Flash, eMMC, Ethernet (PHY integration, bare-metal PS Ethernet), SPI, UART, I2C, GPIO
- **Embedded Software:** C (primary, bare-metal and embedded), C++ (working knowledge), PetaLinux
- **Scripting & Tools:** Python, TCL (Vivado/Vitis automation), Git (currently adopting for daily version control)
- **Hardware Design:** Schematic design for a custom FPGA board, high-speed signal integrity considerations

WORK EXPERIENCE

Hardware & Firmware Engineer, Guided Systems Division — DSO National Laboratories *Jul 2024 – Present*

- Architected and implemented VHDL RTL for a continuous, high-throughput data acquisition and DSP streaming pipeline on a Xilinx FPGA platform, using AXI4-Stream with buffering and backpressure handling to sustain throughput
- Built AXI DMA data paths moving streamed data into DDR memory for real-time processing, alongside QSPI flash and eMMC storage interfaces
- Closed timing on own designs in Vivado by analysing static timing reports, tracing failing paths, and reworking the VHDL RTL to fix them
- Designed the schematic for a custom Xilinx Zynq-7000 FPGA board with high-speed signal integrity considerations and an onboard Ethernet PHY; brought up the board and implemented bare-metal PS Ethernet at 100Mbps
- Debugged boards and FPGA designs in the lab with oscilloscope and JTAG debugger, and integrated FPGA, embedded C firmware, and PetaLinux components with software and systems engineering teams
- Automated Vivado/Vitis design and build tasks with TCL scripting
- Part of a team of engineers piloting the use of AI in FPGA development, applying AI-assisted workflows to RTL design, debugging, and technical documentation

INTERNSHIP EXPERIENCE

FPGA Intern, Guided Systems Division — DSO National Laboratories *May 2023 – Aug 2023*

- Implemented a real-time image processing algorithm (CLAHE) with Vivado HLS on a Xilinx Zynq UltraScale+ platform, optimising to meet real-time throughput and latency requirements
- Wrote C to read from and write to DDR3 memory as part of the processing pipeline

FPGA Intern, Guided Systems Division — DSO National Laboratories *May 2022 – Dec 2022*

- Designed VHDL RTL for raw data flow and interfacing on an FMCW radar system with FPGA and radar engineers, from architecture through field trials
- Worked with AXI4-Stream, UART, RS422, and FFT-based signal processing; processed collected radar data into Synthetic Aperture Radar (SAR) images

Hardware Testing Intern, Guided Systems Division — DSO National Laboratories *Oct 2018 – Mar 2019*

- Tested and validated VITA 74 processor cards across USB, UART, SPI, PCIe, CAN Bus, and I2C, characterising CPU, GPU, SSD, and USB performance on a test jig

EDUCATION

Nanyang Technological University, Singapore *Aug 2021 – Aug 2024*

B.Eng. Electrical & Electronic Engineering — Honours (Distinction), Specialised in Info-communications

Temasek Polytechnic, Singapore *Apr 2016 – May 2019*

Diploma in Electronics (CGPA 3.90/4.00) — Diploma with Merit

AWARDS & ACHIEVEMENTS

- DSO SOAR Scholarship (2022–2023); DSO Diploma Scholarship (2017–2019)
- Cohort Silver Medalist, Temasek Polytechnic Diploma in Electronics (2018/19)
- Bronze, WorldSkills Singapore 2018 — Industrial Electronics; Gold, Temasek Polytechnic Electronic Design Competition (2018)
- Champion Team, NTU Design & Innovation Project Competition (2023) — Team Leader

COURSES & TRAINING

- Petalinux for AMD FPGA; Linux for Altera FPGA

TECHNICAL INTERESTS

- FPGA Acceleration, Low-latency Systems, Computer Architecture, Computer Networking, Modern C++, Linux Systems Programming